

SAT Math Mastery · Week 2A

Linear Systems

Time	150 minutes: 110 min solving + 40 min self-check and error log
Topics	number of solutions · the “second equation” · target expressions without solving · solutions and constants · parametric common points · graphs of systems · elimination shortcuts · category, mixture and rate systems
Name / Date	_____

1 · SKILLS & IDEAS

- S1.** Decide whether a 2×2 linear system has one, none or infinitely many solutions from its coefficients.
- S2.** Write or choose a second equation that forces a required number of solutions.
- S3.** Find a target expression ($x + y$, $2x - y$, $3x + 3y \dots$) by adding or subtracting equations, without solving for x and y .
- S4.** Use a given solution or intersection point to find unknown constants in the equations.
- S5.** Find the point shared by a whole family of lines ℓ_t by making the coefficient of the parameter vanish.
- S6.** Read a system from a graph (slopes, intercepts, intersection) and use it with a parameter.
- S7.** Choose the fastest elimination: add, subtract, scale, or use symmetry in the coefficients.
- S8.** Model counts, prices, mixtures, ratios and rate–time situations as a system and answer the quantity asked.

2 · REQUIRED FACTS

- The solution of a system is the ordered pair that makes *both* equations true; on a graph it is the intersection point.
- Two lines with different slopes meet exactly once; equal slopes with different intercepts never meet; equal slopes with equal intercepts are the same line.
- A system has infinitely many solutions only if one equation is a nonzero multiple of the other (every part, including the constant).
- Scaling one equation changes nothing; scaling only the left side and changing the constant creates a parallel line.
- A line $y = mx + b$ written as $Ax + By = C$ has $m = -\frac{A}{B}$; compare slopes before comparing constants.
- A problem often asks for an expression such as $x + y$ or a single variable: read the question before solving.
- Time = $\frac{\text{distance}}{\text{rate}}$; convert minutes to hours (or hours to minutes) before substituting.

3 · REQUIRED FORMULAS

$$\begin{cases} a_1x + b_1y = c_1 \\ a_2x + b_2y = c_2 \end{cases} \quad \text{one solution: } \frac{a_1}{a_2} \neq \frac{b_1}{b_2}; \quad \text{none: } \frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}; \quad \text{infinite: } \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

$$\text{Cramer's rule (check only): } x = \frac{c_1b_2 - c_2b_1}{a_1b_2 - a_2b_1}, \quad y = \frac{a_1c_2 - a_2c_1}{a_1b_2 - a_2b_1}$$

$$\text{Parametric family: } t(\text{expression}_1) + (\text{expression}_2) = 0 \Rightarrow \text{expression}_1 = 0, \text{ expression}_2 = 0 \quad d = rt, \quad t = \frac{d}{r}$$

4 · MINI THEORY SHEET

Elimination vs. substitution.

Look for	Move
a variable already isolated ($y = 2x - 1$)	substitute into the other equation
equal or opposite coefficients	add or subtract the equations
coefficients that are multiples of each other	scale one equation, then add or subtract
a question about $x + y$, $x - y$, $2x + 2y$	combine the equations to produce that expression directly
a graph with two lines	read two lattice points on each line, rebuild the equations, then solve

Number of solutions in one line. Compare $\frac{a_1}{a_2}$, $\frac{b_1}{b_2}$, $\frac{c_1}{c_2}$. Cross-multiplying is safer than dividing: $a_1b_2 = a_2b_1$ means parallel.

The second equation. For no solution, keep the left side (or a multiple of it) and use a constant that is *not* the matching multiple. For infinitely many, multiply the whole equation. For exactly one, any equation whose coefficient ratio differs.

Parameter conditions. Turn the requirement (no solution, infinite solutions, given point) into an equation in the constant; then check the constants too, because a value that makes the lines parallel may also make them identical.

Word systems. Name the unknowns in the units the question uses. A count equation and a value (or rate) equation are the usual pair; for ratio chains, express every count with one variable.

Speed shortcuts.

Situation	Shortcut
$x + y$ wanted, coefficients swap ($4x + y$, $x + 4y$)	add: $5(x + y) = \text{sum}$
$x - y$ wanted, same swap	subtract the equations
target is $px + qy$	find u, v with $u(\text{eq}_1) + v(\text{eq}_2) = px + qy$ (match both coefficients)
solution (x_0, y_0) given, constants unknown	substitute, then add/subtract the two new equations
parametric family	write $t(\dots) + (\dots) = 0$ and solve the two brackets
parallel test	$a_1b_2 = a_2b_1$; then compare the constants
mixture of two prices	total value – (all cheaper item) = (price gap) $\times$ (count of the dearer item)
two speeds, one total time	use $t = \frac{d}{r}$ for each leg, then add the times

5 · GUIDED EXAMPLES

Example 1 — Number of solutions and the second equation

The first equation is $4x - 6y = 9$. Give a second equation so that the system has (a) no solution, (b) infinitely many solutions, (c) exactly one solution.

(a) Same left side (or a multiple), wrong constant: $2x - 3y = 5$ (multiplying by 2 gives $4x - 6y = 10 \neq 9$). (b) Multiply *everything*: $8x - 12y = 18$. (c) Different coefficient ratio: $3x + y = 2$, since $4 \cdot 1 \neq (-6) \cdot 3$.

Example 2 — Target expression without solving

If $5x + 3y = 21$ and $3x + 5y = 19$, find $x + y$ and $x - y$.

Add: $8x + 8y = 40 \Rightarrow x + y = 5$. Subtract: $2x - 2y = 2 \Rightarrow x - y = 1$. (Check: $x = 3$, $y = 2$ works in both equations.)

Example 3 — Common point of a family of lines

For every real t the line $(t + 1)x + (2 - t)y = 3t + 6$ passes through one fixed point. Find it.

Collect the t terms: $t(x - y - 3) + (x + 2y - 6) = 0$. This holds for every t only if $x - y - 3 = 0$ and $x + 2y - 6 = 0$.

Subtract: $3y = 3 \Rightarrow y = 1$, so $x = 4$. The point is $(4, 1)$.

6 · PRACTICE SET — 36 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1.

EASY · NO CALCULATOR NEEDED

What is the value of x in the system of equations below?

$$y = 2x - 1$$

$$3x + y = 19$$

(A) 3

(B) 4 ✓

(C) 7

(D) 10

- Given: $y = 2x - 1$ and $3x + y = 19$. Asked: the value of x .
- Method: substitution, because the first equation already tells us what y equals in terms of x .
- Replace y in the second equation with $2x - 1$: $3x + (2x - 1) = 19$.
- Combine like terms ($3x + 2x = 5x$): $5x - 1 = 19$.
- Add 1 to both sides: $5x - 1 + 1 = 19 + 1$, so $5x = 20$.
- Divide both sides by 5: $x = \frac{20}{5} = 4$.
- Check: $y = 2(4) - 1 = 7$. First equation: $7 = 2(4) - 1 = 7$, true. Second equation: $3(4) + 7 = 12 + 7 = 19$, true. The value $x = 4$ is option (B).
- So the answer is (B) 4.

Answer: (B)

2.

EASY · NO CALCULATOR NEEDED

$$y = 3x - 5$$

$$6x - 2y = 10$$

How many solutions does the given system of equations have?

(A) Exactly two

(B) Zero

(C) Infinitely many ✓

(D) Exactly one

- Given: $y = 3x - 5$ and $6x - 2y = 10$. Asked: how many solutions the system has.
- Method: substitution. Replacing y in the second equation shows whether the two equations agree for every x , for no x , or for exactly one x .
- Replace y with $3x - 5$ in $6x - 2y = 10$: $6x - 2(3x - 5) = 10$.
- Distribute -2 over the bracket: $6x - 6x + 10 = 10$.
- Combine like terms ($6x - 6x = 0$): $10 = 10$.
- The statement $10 = 10$ is true for every value of x , so every point on the line $y = 3x - 5$ satisfies both equations.
- Check: divide $6x - 2y = 10$ by 2 to get $3x - y = 5$, which rearranges to $y = 3x - 5$. Both equations describe the same line. For example, $x = 0$ gives $y = -5$, and $6(0) - 2(-5) = 10$, true.
- So the answer is (C) Infinitely many solutions.

Answer: (C)

3.

EASY · NO CALCULATOR NEEDED

$$4x + y = 17$$

$$x + 4y = 13$$

The system of equations above is satisfied by (x, y) . What is the value of $x + y$?

1. Given: $4x + y = 17$ and $x + 4y = 13$. Asked: the value of $x + y$.
2. Method: add the equations. The coefficients on x and y become equal after adding, which produces a multiple of $x + y$.
3. Add the left sides and the right sides: $(4x + y) + (x + 4y) = 17 + 13$.
4. Combine like terms: $4x + x = 5x$ and $y + 4y = 5y$, and $17 + 13 = 30$. So $5x + 5y = 30$.
5. Factor out 5 on the left: $5(x + y) = 30$.
6. Divide both sides by 5: $x + y = \frac{30}{5} = 6$.
7. Check by finding x and y : from $4x + y = 17$, $y = 17 - 4x$. Substituting into $x + 4y = 13$ gives $x + 4(17 - 4x) = 13$, then $x + 68 - 16x = 13$, then $-15x = -55$, so $x = \frac{11}{3}$ and $y = 17 - \frac{44}{3} = \frac{7}{3}$. Then $x + y = \frac{18}{3} = 6$, and $4\left(\frac{11}{3}\right) + \frac{7}{3} = \frac{51}{3} = 17$, true.
8. So the answer is 6.

Answer: 6

4.

EASY · NO CALCULATOR NEEDED

At a school play, adult tickets cost \$8 each and student tickets cost \$5 each. A total of 120 tickets were sold for \$735. If a is the number of adult tickets and s is the number of student tickets sold, which system of equations represents this situation?

(A) $a + s = 120$ and $8a + 5s = 735$ ✓

(B) $a + s = 735$ and $8a + 5s = 120$

(C) $a + s = 120$ and $5a + 8s = 735$

(D) $a + s = 120$ and $8a - 5s = 735$

1. Given: adult tickets cost 8 dollars, student tickets cost 5 dollars, 120 tickets were sold, and the total was 735 dollars. The number of adult tickets is a and student tickets is s . Asked: the system of equations that matches the story.
2. Method: write one equation for each fact, one for the number of tickets and one for the money collected.
3. Number of tickets: adult tickets plus student tickets is 120, so $a + s = 120$.
4. Money: adult tickets bring in $8 \cdot a = 8a$ dollars and student tickets bring in $5 \cdot s = 5s$ dollars, and together they bring in 735 dollars, so $8a + 5s = 735$.
5. The system $a + s = 120$ and $8a + 5s = 735$ is option (A).
6. Check with $a = 45$ and $s = 75$: $45 + 75 = 120$. Money: $8(45) = 360$ and $5(75) = 375$, and $360 + 375 = 735$. Both equations hold.
7. So the answer is (A) $a + s = 120$ and $8a + 5s = 735$.

Answer: (A)

5.

EASY · NO CALCULATOR NEEDED

$$2x + y = 11$$

$$kx - y = 4$$

In the system of equations above, k is a constant. If $(3, 5)$ is the solution to the system, what is the value of k ?

(A) $-\frac{1}{3}$

(B) 1

(C) $\frac{4}{3}$

(D) 3 ✓

- Given: $2x + y = 11$ and $kx - y = 4$, with solution $(3, 5)$, so $x = 3$ and $y = 5$. Asked: the value of k .
- Method: substitute the known solution into the equation that contains k . First confirm the point works in the equation without k .
- Check the first equation: $2(3) + 5 = 6 + 5 = 11$, true.
- Substitute $x = 3$ and $y = 5$ into $kx - y = 4$: $k(3) - 5 = 4$, which is $3k - 5 = 4$.
- Add 5 to both sides: $3k = 9$.
- Divide both sides by 3: $k = 3$.
- Check: $3(3) - 5 = 9 - 5 = 4$, true. The value $k = 3$ is option (D).
- So the answer is (D) 3.

Answer: (D)

6.

MEDIUM · NO CALCULATOR NEEDED

$$3x - 4y = 10$$

One equation of a system of two linear equations is shown. The system has no solution. Which of the following could be the second equation?

(A) $3x + 4y = 10$

(B) $6x - 8y = 20$

(C) $6x - 8y = 15$ ✓

(D) $8x - 6y = 20$

- Given: one equation is $3x - 4y = 10$, and the system has no solution. Asked: which option could be the second equation.
- Method: slope comparison. A system has no solution when the two lines are parallel and different, meaning same slope and different y -intercept. Write each equation as $y = mx + b$.
- First equation: subtract $3x$ from both sides, $-4y = -3x + 10$. Divide by -4 : $y = \frac{3}{4}x - \frac{5}{2}$. Slope $\frac{3}{4}$, intercept $-\frac{5}{2}$.
- (A) $3x + 4y = 10$: subtract $3x$, $4y = -3x + 10$. Divide by 4: $y = -\frac{3}{4}x + \frac{5}{2}$. The slope is $-\frac{3}{4}$, different from $\frac{3}{4}$, so the lines cross once.
- (B) $6x - 8y = 20$: subtract $6x$, $-8y = -6x + 20$. Divide by -8 : $y = \frac{3}{4}x - \frac{5}{2}$. Same slope and same intercept, so it is the same line (infinitely many solutions).
- (C) $6x - 8y = 15$: subtract $6x$, $-8y = -6x + 15$. Divide by -8 : $y = \frac{3}{4}x - \frac{15}{8}$. The slope is $\frac{3}{4}$, and the intercept $-\frac{15}{8}$ differs from $-\frac{5}{2} = -\frac{20}{8}$. Parallel and different lines, so no solution.
- (D) $8x - 6y = 20$: subtract $8x$, $-6y = -8x + 20$. Divide by -6 : $y = \frac{4}{3}x - \frac{10}{3}$. The slope $\frac{4}{3}$ differs from $\frac{3}{4}$, so the lines cross once.
- Check (C): doubling every term of $3x - 4y = 10$ gives $6x - 8y = 20$. Option (C) says the same left side equals 15, and $20 \neq 15$, so no pair (x, y) can satisfy both.
- So the answer is (C) $6x - 8y = 15$.

Answer: (C)

7.

MEDIUM · NO CALCULATOR NEEDED

Which of the following systems of linear equations has infinitely many solutions?

(A) $2x - 5y = 10$

$6x - 15y = 20$

(B) $4x + 6y = 8$ ✓

$6x + 9y = 12$

(C) $x - 3y = 4$

$2x - 6y = 9$

(D) $3x + y = 7$

$6x + 3y = 14$

- Given: four systems of two equations each. Asked: which system has infinitely many solutions.
- Method: ratio comparison. Two equations $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$ describe the same line when $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$.
- (A) $2x - 5y = 10$ and $6x - 15y = 20$: $\frac{2}{6} = \frac{1}{3}$, $\frac{-5}{-15} = \frac{1}{3}$, and $\frac{10}{20} = \frac{1}{2}$. The third ratio is different, so the lines are parallel and different.
- (B) $4x + 6y = 8$ and $6x + 9y = 12$: $\frac{4}{6} = \frac{2}{3}$, $\frac{6}{9} = \frac{2}{3}$, and $\frac{8}{12} = \frac{2}{3}$. All three ratios are equal.
- Confirm (B): multiply the first equation by $\frac{3}{2}$. $\frac{3}{2}(4x) = 6x$, $\frac{3}{2}(6y) = 9y$, and $\frac{3}{2}(8) = 12$, giving $6x + 9y = 12$, which is exactly the second equation.
- (C) $x - 3y = 4$ and $2x - 6y = 9$: $\frac{1}{2}$, $\frac{-3}{-6} = \frac{1}{2}$, and $\frac{4}{9}$. Since $\frac{4}{9} \neq \frac{1}{2}$, the lines are parallel and different.
- (D) $3x + y = 7$ and $6x + 3y = 14$: $\frac{3}{6} = \frac{1}{2}$ and $\frac{1}{3}$. These differ, so the lines cross once.
- Only system (B) has all three ratios equal.
- So the answer is (B) the system $4x + 6y = 8$ and $6x + 9y = 12$.

Answer: (B)

8.

MEDIUM · NO CALCULATOR NEEDED

$x - 4y = 7$

$3x + 2y = 7$

If (x, y) is the solution to the system of equations above, what is the value of $4x - 2y$?

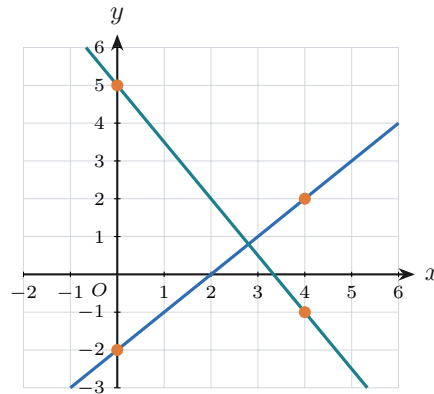
- Given: $x - 4y = 7$ and $3x + 2y = 7$. Asked: the value of $4x - 2y$.
- Method: add the equations. Adding gives $x + 3x = 4x$ and $-4y + 2y = -2y$, which is exactly the expression we want.
- Add the left sides: $(x - 4y) + (3x + 2y) = (x + 3x) + (-4y + 2y) = 4x - 2y$.
- Add the right sides: $7 + 7 = 14$.
- So $4x - 2y = 14$.
- Check by solving the system: multiply $3x + 2y = 7$ by 2 to get $6x + 4y = 14$. Add this to $x - 4y = 7$: $7x = 21$, so $x = 3$.
- Substitute $x = 3$ into $3x + 2y = 7$: $9 + 2y = 7$. Subtract 9: $2y = -2$. Divide by 2: $y = -1$.
- Check both original equations: $3 - 4(-1) = 3 + 4 = 7$ and $3(3) + 2(-1) = 9 - 2 = 7$. Then $4(3) - 2(-1) = 12 + 2 = 14$.
- So the answer is 14.

Answer: 14

9.

MEDIUM · DESMOS

The graph of a system of two linear equations in the xy -plane is shown. The system has exactly one solution (x, y) . What is the value of y ?



- (A) $\frac{7}{5}$
(C) $\frac{16}{5}$

- (B) $\frac{14}{5}$
(D) $\frac{4}{5}$ ✓

- Given (read from the existing steps): the first line passes through $(0, -2)$ and $(4, 2)$, and the second line passes through $(0, 5)$ and $(4, -1)$. Asked: the y -value where they meet.
- Method: write each line as $y = mx + b$ and set the two expressions equal.
- First line slope: $\frac{2 - (-2)}{4 - 0} = \frac{4}{4} = 1$. The y -intercept is -2 , so $y = x - 2$.
- Second line slope: $\frac{-1 - 5}{4 - 0} = \frac{-6}{4} = -\frac{3}{2}$. The y -intercept is 5 , so $y = -\frac{3}{2}x + 5$.
- Set the two expressions for y equal: $x - 2 = -\frac{3}{2}x + 5$.
- Add $\frac{3}{2}x$ to both sides ($x + \frac{3}{2}x = \frac{5}{2}x$): $\frac{5}{2}x - 2 = 5$. Then add 2 to both sides: $\frac{5}{2}x = 7$.
- Multiply both sides by $\frac{2}{5}$: $x = 7 \cdot \frac{2}{5} = \frac{14}{5}$.
- Substitute into $y = x - 2$: $y = \frac{14}{5} - 2 = \frac{14}{5} - \frac{10}{5} = \frac{4}{5}$.
- Check in the second line: $-\frac{3}{2} \cdot \frac{14}{5} + 5 = -\frac{21}{5} + \frac{25}{5} = \frac{4}{5}$, which agrees. The value $y = \frac{4}{5}$ is option (D).
- So the answer is (D) $\frac{4}{5}$.

Answer: (D)

10.

MEDIUM · NO CALCULATOR NEEDED

$$2x + ky = 8$$

$$6x + 9y = 20$$

In the system of equations above, k is a constant. If the system has no solution, what is the value of k ?

- Given: $2x + ky = 8$ and $6x + 9y = 20$, and the system has no solution. Asked: the value of k .
- Method: slope comparison. No solution means the lines are parallel and different, so the slopes must be equal.
- Second equation: subtract $6x$, $9y = -6x + 20$. Divide by 9 : $y = -\frac{2}{3}x + \frac{20}{9}$. Slope $-\frac{2}{3}$.
- First equation: subtract $2x$, $ky = -2x + 8$. Divide by k : $y = -\frac{2}{k}x + \frac{8}{k}$. Slope $-\frac{2}{k}$.
- Set the slopes equal: $-\frac{2}{k} = -\frac{2}{3}$, which gives $\frac{2}{k} = \frac{2}{3}$.
- Cross-multiply: $2 \cdot 3 = 2 \cdot k$, so $6 = 2k$. Divide by 2 : $k = 3$.
- Check the lines are different: with $k = 3$ the first equation is $2x + 3y = 8$. Multiply by 3 : $6x + 9y = 24$. The second equation says $6x + 9y = 20$, and $24 \neq 20$, so no pair (x, y) satisfies both.
- So the answer is 3 .

Answer: 3

11.

MEDIUM · NO CALCULATOR NEEDED

A collection of 84 stamps contains only domestic, regional and international stamps. There are twice as many regional stamps as domestic stamps, and there are 6 fewer international stamps than regional stamps. How many international stamps are in the collection?

(A) 30 ✓

(B) 36

(C) 18

(D) 24

- Given: 84 stamps in total. Regional stamps are twice the domestic stamps. International stamps are 6 fewer than regional stamps. Asked: the number of international stamps.
- Method: let one letter stand for the domestic stamps, write the other two counts in terms of it, and add them to make 84.
- Let d be the number of domestic stamps. Then regional = $2d$ and international = $2d - 6$.
- Total: $d + 2d + (2d - 6) = 84$.
- Combine like terms ($d + 2d + 2d = 5d$): $5d - 6 = 84$.
- Add 6 to both sides: $5d = 90$.
- Divide both sides by 5: $d = 18$.
- International stamps: $2d - 6 = 2(18) - 6 = 36 - 6 = 30$.
- Check: regional = $2(18) = 36$, and $18 + 36 + 30 = 84$. Also $36 - 6 = 30$. The number 30 is option (A).
- So the answer is (A) 30.

Answer: (A)

12.

MEDIUM · CALCULATOR

A coffee shop sold 64 cups of coffee one morning. Small cups cost \$3 each and large cups cost \$5 each, and the total sales were \$270. How many large cups were sold?

- Given: s small cups at 3 dollars and ℓ large cups at 5 dollars. Cups: $s + \ell = 64$. Sales: $3s + 5\ell = 270$. Asked: ℓ .
- Method: substitution. The first equation is easy to solve for s .
- Subtract ℓ from both sides of $s + \ell = 64$: $s = 64 - \ell$.
- Substitute into $3s + 5\ell = 270$: $3(64 - \ell) + 5\ell = 270$.
- Distribute 3: $192 - 3\ell + 5\ell = 270$.
- Combine like terms ($-3\ell + 5\ell = 2\ell$): $192 + 2\ell = 270$.
- Subtract 192 from both sides: $2\ell = 78$.
- Divide both sides by 2: $\ell = 39$.
- Then $s = 64 - 39 = 25$. Check: $25 + 39 = 64$, and $3(25) + 5(39) = 75 + 195 = 270$, both true.
- So the answer is 39.

Answer: 39

13.

MEDIUM · CALCULATOR

Leila drives from home to a lake at an average speed of 60 miles per hour and returns by the same road at an average speed of 40 miles per hour. Her total driving time for the round trip is 5 hours. How many miles is the lake from her home?

- Given: speed out 60 miles per hour, speed back 40 miles per hour, total time 5 hours. Asked: the one-way distance d in miles.
- Method: time equals distance divided by speed. Write the two times, add them to make 5, then clear the fractions.
- Time going: $\frac{d}{60}$ hours. Time returning: $\frac{d}{40}$ hours.
- Total time: $\frac{d}{60} + \frac{d}{40} = 5$.
- The common denominator of 60 and 40 is 120. Multiply every term by 120: $120 \cdot \frac{d}{60} + 120 \cdot \frac{d}{40} = 120 \cdot 5$.
- Simplify each term: $2d + 3d = 600$.
- Combine like terms: $5d = 600$.
- Divide both sides by 5: $d = 120$.
- Check: going takes $\frac{120}{60} = 2$ hours, returning takes $\frac{120}{40} = 3$ hours, and $2 + 3 = 5$ hours.
- So the answer is 120.

Answer: 120

14.

HARD · NO CALCULATOR NEEDED

$$6x + 9y = 21$$

One equation of a system of two linear equations is shown. The system has exactly one solution. Which of the following could be the second equation?

(A) $2x + 3y = 11$

(B) $9x + 6y = 21$ ✓

(C) $-4x - 6y = -14$

(D) $y = -\frac{2}{3}x + \frac{7}{3}$

- Given: one equation is $6x + 9y = 21$, and the system has exactly one solution. Asked: which option could be the second equation.
- Method: slope comparison. Exactly one solution means the two lines have different slopes. Write each equation as $y = mx + b$.
- Given equation: subtract $6x$, $9y = -6x + 21$. Divide by 9: $y = -\frac{2}{3}x + \frac{7}{3}$. Slope $-\frac{2}{3}$, intercept $\frac{7}{3}$.
- (A) $2x + 3y = 11$: subtract $2x$, $3y = -2x + 11$. Divide by 3: $y = -\frac{2}{3}x + \frac{11}{3}$. Same slope, different intercept, so no solution.
- (B) $9x + 6y = 21$: subtract $9x$, $6y = -9x + 21$. Divide by 6: $y = -\frac{3}{2}x + \frac{7}{2}$. The slope $-\frac{3}{2}$ differs from $-\frac{2}{3}$, so the lines cross once.
- (C) $-4x - 6y = -14$: add $4x$, $-6y = 4x - 14$. Divide by -6 : $y = -\frac{2}{3}x + \frac{7}{3}$. Same slope and same intercept, so it is the same line.
- (D) $y = -\frac{2}{3}x + \frac{7}{3}$ is already in slope-intercept form and matches the given line exactly, so it is the same line.
- Only option (B) has a different slope.
- Check by solving: divide $6x + 9y = 21$ by 3 to get $2x + 3y = 7$, and multiply by 2 to get $4x + 6y = 14$. The equation $9x + 6y = 21$ has the same y -term, so subtract: $(9x - 4x) + (6y - 6y) = 21 - 14$, which gives $5x = 7$ and $x = \frac{7}{5}$.
- Substitute $x = \frac{7}{5}$ into $2x + 3y = 7$: $\frac{14}{5} + 3y = 7$. Subtract $\frac{14}{5}$: $3y = \frac{21}{5}$. Divide by 3: $y = \frac{7}{5}$.
- Check both equations at $\left(\frac{7}{5}, \frac{7}{5}\right)$: $6 \cdot \frac{7}{5} + 9 \cdot \frac{7}{5} = \frac{105}{5} = 21$ and $9 \cdot \frac{7}{5} + 6 \cdot \frac{7}{5} = \frac{105}{5} = 21$. There is exactly one solution.
- So the answer is (B) $9x + 6y = 21$.

Answer: (B)

15.

HARD · NO CALCULATOR NEEDED

$$ax + by = 13$$

$$bx + ay = 12$$

In the system above, a and b are constants. If $(3, 2)$ is the solution to the system, what is the value of $a^2 - b^2$?

- Given: $ax + by = 13$ and $bx + ay = 12$, with solution $(3, 2)$. Asked: the value of $a^2 - b^2$.
- Method: substitute the point to get two equations in a and b , then use $a^2 - b^2 = (a - b)(a + b)$.
- First equation with $x = 3$, $y = 2$: $a(3) + b(2) = 13$, which is $3a + 2b = 13$.
- Second equation with $x = 3$, $y = 2$: $b(3) + a(2) = 12$, which is $2a + 3b = 12$.
- Subtract the second from the first: $(3a - 2a) + (2b - 3b) = 13 - 12$, so $a - b = 1$.
- Add the two equations: $(3a + 2a) + (2b + 3b) = 13 + 12$, so $5a + 5b = 25$.
- Divide both sides by 5: $a + b = 5$.
- Factor the difference of squares: $a^2 - b^2 = (a - b)(a + b)$.
- Substitute: $(a - b)(a + b) = 1 \cdot 5 = 5$.
- Check by finding a and b : adding $a - b = 1$ and $a + b = 5$ gives $2a = 6$, so $a = 3$ and $b = 2$. Then $3(3) + 2(2) = 13$ and $2(3) + 3(2) = 12$, both true, and $9 - 4 = 5$.
- So the answer is 5.

Answer: 5

16.

HARD · NO CALCULATOR NEEDED

$$3x + y = 7$$

$$x + 3y = 9$$

If (x, y) is the solution to the system of equations above, what is the value of $5x - y$?

1. Given: $3x + y = 7$ and $x + 3y = 9$. Asked: the value of $5x - y$.
2. Method: elimination to find x and y , then evaluate $5x - y$.
3. Multiply every term of $x + 3y = 9$ by 3: $3x + 9y = 27$.
4. Subtract this from $3x + y = 7$: $(3x - 3x) + (y - 9y) = 7 - 27$, so $-8y = -20$.
5. Divide both sides by -8 : $y = \frac{20}{8} = \frac{5}{2}$.
6. Substitute $y = \frac{5}{2}$ into $3x + y = 7$: $3x + \frac{5}{2} = 7$.
7. Subtract $\frac{5}{2}$ from both sides: $3x = 7 - \frac{5}{2} = \frac{14}{2} - \frac{5}{2} = \frac{9}{2}$.
8. Divide both sides by 3: $x = \frac{9}{2} \cdot \frac{1}{3} = \frac{3}{2}$.
9. Evaluate: $5x - y = 5 \cdot \frac{3}{2} - \frac{5}{2} = \frac{15}{2} - \frac{5}{2} = \frac{10}{2} = 5$.
10. Check the original equations: $3 \cdot \frac{3}{2} + \frac{5}{2} = \frac{9}{2} + \frac{5}{2} = 7$ and $\frac{3}{2} + 3 \cdot \frac{5}{2} = \frac{3}{2} + \frac{15}{2} = 9$, both true.
11. Cross-check: $2(3x + y) - (x + 3y) = 6x + 2y - x - 3y = 5x - y$, and $2(7) - 9 = 14 - 9 = 5$.
12. So the answer is 5.

Answer: 5

17.

HARD · NO CALCULATOR NEEDED

For each real number t , line ℓ_t in the xy -plane is defined by $y = (t + 3)x - 2t + 1$. Which of the following points lies on ℓ_t for every value of t ?

(A) (7, 2)

(B) (1, 4)

(C) (2, 7) ✓

(D) (2, 5)

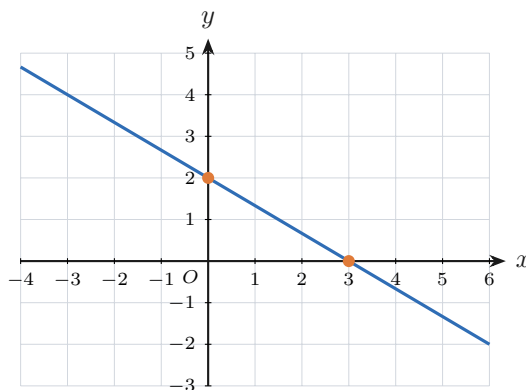
1. Given: for every real number t , the line is $y = (t + 3)x - 2t + 1$. Asked: which point lies on the line for every t .
2. Method: regroup the equation by powers of t . A point that works for every t cannot depend on t , so the coefficient of t must be zero.
3. Distribute x : $y = tx + 3x - 2t + 1$.
4. Group the terms that contain t : $tx - 2t = t(x - 2)$. So $y = t(x - 2) + 3x + 1$.
5. For y to give the same value no matter what t is, the part multiplying t must be 0: $x - 2 = 0$.
6. Add 2 to both sides: $x = 2$.
7. With $x = 2$ the t term disappears: $y = t(0) + 3(2) + 1 = 6 + 1 = 7$. The point is (2, 7).
8. Check in the original equation with $x = 2$: $y = (t + 3)(2) - 2t + 1 = 2t + 6 - 2t + 1 = 7$, for any t .
9. Numerical check: $t = 0$ gives $y = 3x + 1$, and at $x = 2$ that is 7. $t = 1$ gives $y = 4x - 1$, and at $x = 2$ that is 7.
10. The point (2, 7) is option (C).
11. So the answer is (C) (2, 7).

Answer: (C)

18.

HARD · DESMOS

Line ℓ is shown in the xy -plane. The system consisting of line ℓ and the line $kx + 6y = 5$, where k is a constant, has exactly one solution. Which of the following could NOT be the value of k ?



(A) 4 ✓

(B) 5

(C) -2

(D) 1

1. Given (read from the existing steps): line ℓ passes through $(0, 2)$ and $(3, 0)$. The second line is $kx + 6y = 5$. The system has exactly one solution. Asked: which listed value could NOT be k .
2. Method: slope comparison. Exactly one solution requires different slopes, so we look for the value of k that makes the slopes equal.
3. Slope of ℓ : $\frac{0 - 2}{3 - 0} = -\frac{2}{3}$. The y -intercept is 2, so $y = -\frac{2}{3}x + 2$.
4. Slope of $kx + 6y = 5$: subtract kx , $6y = -kx + 5$. Divide by 6: $y = -\frac{k}{6}x + \frac{5}{6}$. The slope is $-\frac{k}{6}$.
5. Set the slopes equal: $-\frac{k}{6} = -\frac{2}{3}$, so $\frac{k}{6} = \frac{2}{3}$. Multiply both sides by 6: $k = 4$.
6. For $k = 4$ the second line is $4x + 6y = 5$. The y -intercepts are 2 and $\frac{5}{6}$, which differ, so the lines are parallel and different: no solution.
7. Equation check: ℓ is $2x + 3y = 6$ (test $(0, 2)$: 6, and $(3, 0)$: 6). Doubling gives $4x + 6y = 12$, and $12 \neq 5$.
8. For $k = 5$ the slope is $-\frac{5}{6}$, for $k = -2$ it is $\frac{1}{3}$, and for $k = 1$ it is $-\frac{1}{6}$. Each differs from $-\frac{2}{3}$, so each gives exactly one solution.
9. Only $k = 4$ fails to give exactly one solution, so k could NOT be 4.
10. So the answer is (A) 4.

Answer: (A)

19.

HARD · CALCULATOR

Priya practices by cycling at a constant speed of 15 miles per hour and by running at a constant speed of 6 miles per hour. In one session she covers a total of 27 miles in 2 hours and 30 minutes. How many minutes did she spend running?

1. Given: cycling speed 15 miles per hour, running speed 6 miles per hour, total distance 27 miles, total time 2 hours 30 minutes = 2.5 hours. Asked: the minutes spent running.
2. Method: let c be the hours cycling and r the hours running, write two equations, and use substitution.
3. Time equation: $c + r = 2.5$. Distance equation: $15c + 6r = 27$.
4. Solve the time equation for c by subtracting r : $c = 2.5 - r$.
5. Substitute into the distance equation: $15(2.5 - r) + 6r = 27$.
6. Distribute 15: $37.5 - 15r + 6r = 27$.
7. Combine like terms ($-15r + 6r = -9r$): $37.5 - 9r = 27$.
8. Subtract 37.5 from both sides: $-9r = -10.5$.
9. Divide both sides by -9 : $r = \frac{10.5}{9} = \frac{21}{18} = \frac{7}{6}$ hours.
10. Convert to minutes: $\frac{7}{6} \times 60 = 70$ minutes.
11. Check: $c = 2.5 - \frac{7}{6} = \frac{15}{6} - \frac{7}{6} = \frac{8}{6} = \frac{4}{3}$ hours. Time: $\frac{4}{3} + \frac{7}{6} = \frac{8}{6} + \frac{7}{6} = \frac{15}{6} = 2.5$. Distance: $15 \cdot \frac{4}{3} = 20$ and $6 \cdot \frac{7}{6} = 7$, and $20 + 7 = 27$.
12. So the answer is 70.

Answer: 70**20.**

HARD · CALCULATOR

A metalworker melts together an alloy that is 20% silver and an alloy that is 50% silver to make 12 kilograms of a new alloy that is 26% silver. How many kilograms of the 20% silver alloy are used?

(A) 2.4 (B) 3.12

(C) 6 (D) 9.6 ✓

1. Given: x kilograms of the 20% alloy and y kilograms of the 50% alloy make 12 kilograms of a 26% alloy. Asked: the value of x .
2. Method: write one equation for the total mass and one for the mass of silver, then use substitution.
3. Total mass: $x + y = 12$.
4. Silver mass: $0.20x + 0.50y = 0.26(12)$.
5. Compute the right side: $0.26 \times 12 = 3.12$, so $0.20x + 0.50y = 3.12$.
6. From the total mass equation, subtract y : $x = 12 - y$.
7. Substitute into the silver equation: $0.20(12 - y) + 0.50y = 3.12$.
8. Distribute 0.20: $2.4 - 0.20y + 0.50y = 3.12$.
9. Combine like terms ($-0.20y + 0.50y = 0.30y$): $2.4 + 0.30y = 3.12$.
10. Subtract 2.4 from both sides: $0.30y = 0.72$.
11. Divide both sides by 0.30: $y = \frac{0.72}{0.30} = 2.4$.
12. Then $x = 12 - 2.4 = 9.6$ kilograms.
13. Check: silver is $0.20(9.6) + 0.50(2.4) = 1.92 + 1.2 = 3.12$ kilograms, and total mass is $9.6 + 2.4 = 12$ kilograms. The value 9.6 is option (D).
14. So the answer is (D) 9.6.

Answer: (D)

21.

HARD · DESMOS

The table shows values of two linear functions, f and g . For which value of x is $f(x) = g(x)$?

x	$f(x)$	$g(x)$
0	5	20
2	11	16
4	17	12

(A) 14

(B) 2

(C) 3 ✓

(D) 4

- Given: the table lists $f(0) = 5$, $f(2) = 11$, $f(4) = 17$ and $g(0) = 20$, $g(2) = 16$, $g(4) = 12$, and both functions are linear. Asked: the x where $f(x) = g(x)$.
- Method: write each function as $mx + b$ using the slope and the value at $x = 0$, then set them equal.
- Slope of f : $\frac{11 - 5}{2 - 0} = \frac{6}{2} = 3$.
- Since $f(0) = 5$, the intercept is 5, so $f(x) = 3x + 5$. Check: $f(4) = 12 + 5 = 17$.
- Slope of g : $\frac{16 - 20}{2 - 0} = \frac{-4}{2} = -2$. Since $g(0) = 20$, $g(x) = -2x + 20$. Check: $g(4) = -8 + 20 = 12$.
- Set $f(x) = g(x)$: $3x + 5 = -2x + 20$.
- Add $2x$ to both sides: $5x + 5 = 20$.
- Subtract 5 from both sides: $5x = 15$.
- Divide both sides by 5: $x = 3$.
- Check: $f(3) = 3(3) + 5 = 14$ and $g(3) = -2(3) + 20 = 14$, so they are equal at $x = 3$.
- The value $x = 3$ is option (C).
- So the answer is (C) 3.

Answer: (C)

22.

HARD · NO CALCULATOR NEEDED

$$3x + 2y = 10$$

$$6x + ky = m$$

In the system above, k and m are constants. Consider the following statements.

- If $k = 4$ and $m = 20$, the system has infinitely many solutions.
- If $k = 4$ and $m = 18$, the system has no solution.
- If $k = 3$ and $m = 20$, the system has no solution.

Which of the statements are true?

(A) I only

(B) II only

(C) I and II only ✓

(D) I, II, and III

- Given: $3x + 2y = 10$ and $6x + ky = m$. Three statements are to be judged. Asked: which are true.
- Method: compare the second equation with 2 times the first. Doubling the first equation gives $6x + 4y = 20$.
- Statement I, $k = 4$, $m = 20$: the second equation is $6x + 4y = 20$, identical to the doubled first equation. They are the same line, so there are infinitely many solutions. True.
- Statement II, $k = 4$, $m = 18$: the second equation is $6x + 4y = 18$. The same expression $6x + 4y$ would have to equal both 20 and 18, which is impossible, so there is no solution. True.
- Statement III, $k = 3$, $m = 20$: write both in slope-intercept form. First: $2y = -3x + 10$, so $y = -\frac{3}{2}x + 5$. Second: $3y = -6x + 20$, so $y = -2x + \frac{20}{3}$.
- The slopes $-\frac{3}{2}$ and -2 are different, so the lines cross exactly once. The system has one solution, not none. False.
- Check III by solving: subtract $6x + 3y = 20$ from $6x + 4y = 20$ to get $y = 0$. Then $3x = 10$, so $x = \frac{10}{3}$, one solution.
- So the answer is (C) I and II only.

Answer: (C)

23.

HARD · NO CALCULATOR NEEDED

$$\frac{1}{2}x + \frac{1}{3}y = 7$$

$$\frac{1}{3}x + \frac{1}{2}y = 8$$

If (x, y) is the solution to the system of equations above, what is the value of $x + y$?

1. Given: $\frac{1}{2}x + \frac{1}{3}y = 7$ and $\frac{1}{3}x + \frac{1}{2}y = 8$. Asked: the value of $x + y$.
2. Method: add the equations. The coefficients of x and y are swapped between the equations, so the sum has the same coefficient on both.
3. Add the left sides, grouping like terms: $\left(\frac{1}{2}x + \frac{1}{3}x\right) + \left(\frac{1}{3}y + \frac{1}{2}y\right)$.
4. Add the fractions: $\frac{1}{2} + \frac{1}{3} = \frac{3}{6} + \frac{2}{6} = \frac{5}{6}$. So the left side is $\frac{5}{6}x + \frac{5}{6}y$.
5. Add the right sides: $7 + 8 = 15$. So $\frac{5}{6}x + \frac{5}{6}y = 15$.
6. Factor: $\frac{5}{6}(x + y) = 15$.
7. Multiply both sides by $\frac{6}{5}$: $x + y = 15 \cdot \frac{6}{5} = \frac{90}{5} = 18$.
8. Check by solving: multiply the first equation by 6 to get $3x + 2y = 42$, and the second by 6 to get $2x + 3y = 48$. Multiply the first by 3 and the second by 2: $9x + 6y = 126$ and $4x + 6y = 96$. Subtract: $5x = 30$, so $x = 6$.
9. Then $3(6) + 2y = 42$, so $2y = 24$ and $y = 12$. Check: $\frac{1}{2}(6) + \frac{1}{3}(12) = 3 + 4 = 7$ and $\frac{1}{3}(6) + \frac{1}{2}(12) = 2 + 6 = 8$.
10. Therefore $x + y = 6 + 12 = 18$.
11. So the answer is 18.

Answer: 18

24.

SAT CHALLENGE · DESMOS

For each real number t , line ℓ_t in the xy -plane is defined by $(t + 2)x + (1 - t)y = 4t + 5$. All of these lines pass through the same point P . Line m has equation $y = kx - 5$, where k is a constant, and passes through P . What is the value of k ?

1. Given: the line $(t + 2)x + (1 - t)y = 4t + 5$ for every real t ; every one of these lines passes through the same point P ; line m is $y = kx - 5$ and passes through P . Asked: the value of k .
2. Method: since P lies on the line for every t , collect the terms by t and require that the part multiplying t and the part without t each equal zero.
3. Distribute: $tx + 2x + y - ty = 4t + 5$.
4. Group: the terms containing t are $tx - ty = t(x - y)$, and the others are $2x + y$. So $t(x - y) + (2x + y) = 4t + 5$.
5. Subtract $4t$ and 5 from both sides: $t(x - y) - 4t + (2x + y) - 5 = 0$, that is $t(x - y - 4) + (2x + y - 5) = 0$.
6. This must be 0 for every t . At $t = 0$ it says $2x + y - 5 = 0$; then at $t = 1$ it says $x - y - 4 = 0$. So both brackets are zero.
7. The two equations are $x - y = 4$ and $2x + y = 5$.
8. Add the equations: $(x + 2x) + (-y + y) = 4 + 5$, so $3x = 9$.
9. Divide by 3: $x = 3$.
10. Substitute into $x - y = 4$: $3 - y = 4$. Subtract 3: $-y = 1$. So $y = -1$. Then $P = (3, -1)$.
11. Check P in the original family: for $t = 0$, $2(3) + (-1) = 5 = 4(0) + 5$. For $t = 1$, $3(3) + 0 = 9 = 4(1) + 5$. Both true.
12. Line m passes through P , so substitute $x = 3$, $y = -1$ into $y = kx - 5$: $-1 = k(3) - 5$.
13. Add 5 to both sides: $4 = 3k$. Divide by 3: $k = \frac{4}{3}$.
14. Check: $y = \frac{4}{3}(3) - 5 = 4 - 5 = -1$, which matches P .
15. So the answer is $\frac{4}{3}$.

Answer: $\frac{4}{3}$

25.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$kx + 4y = 8$$

$$9x + ky = 12$$

In the system of equations above, k is a constant. If the system has no solution, what is the value of k ?

(A) -9 **(B) -6 ✓**(C) 6 (D) 12

- Given: $kx + 4y = 8$ and $9x + ky = 12$, and the system has no solution. Asked: the value of k .
- Method: no solution means parallel, different lines. The coefficients must be proportional, and the constants must not follow the same ratio.
- Set the coefficient ratios equal: $\frac{k}{9} = \frac{4}{k}$.
- Cross-multiply: $k \cdot k = 9 \cdot 4$, so $k^2 = 36$.
- Take square roots: $k = 6$ or $k = -6$.
- Test $k = 6$: the equations are $6x + 4y = 8$ and $9x + 6y = 12$.
- Multiply the first by $\frac{3}{2}$: $\frac{3}{2}(6x) = 9x$, $\frac{3}{2}(4y) = 6y$, $\frac{3}{2}(8) = 12$, giving $9x + 6y = 12$. This is the second equation, so the lines are the same and there are infinitely many solutions. $k = 6$ does not work.
- Test $k = -6$: the equations are $-6x + 4y = 8$ and $9x - 6y = 12$.
- Multiply the first by $-\frac{3}{2}$: $-\frac{3}{2}(-6x) = 9x$, $-\frac{3}{2}(4y) = -6y$, $-\frac{3}{2}(8) = -12$, giving $9x - 6y = -12$.
- The second equation says $9x - 6y = 12$. Since $-12 \neq 12$, no pair (x, y) satisfies both: no solution.
- The other options give $k^2 = 81$ for -9 and $k^2 = 144$ for 12 , neither equal to 36 , so the slopes are not equal and each gives one solution.
- So the answer is (B) -6 .

Answer: (B)

26.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$2x + y = k + 4$$

$$x - y = 2k - 1$$

In the system of equations above, k is a constant. The solution (x, y) to the system satisfies $x = 2y$. What is the value of k ?

(A) 2 (B) 5 (C) 0 **(D) 1 ✓**

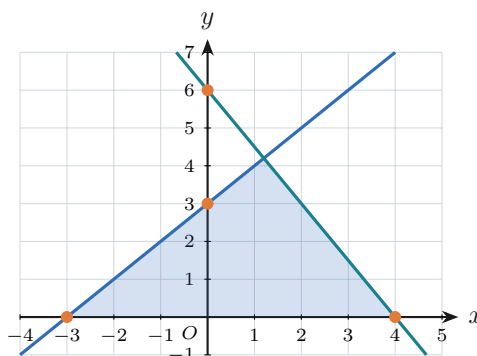
- Given: $2x + y = k + 4$ and $x - y = 2k - 1$, with the solution satisfying $x = 2y$. Asked: the value of k .
- Method: substitution. Replace x by $2y$ in both equations so that only y and k remain.
- First equation: $2(2y) + y = k + 4$, so $4y + y = k + 4$, which is $5y = k + 4$.
- Second equation: $2y - y = 2k - 1$, which is $y = 2k - 1$.
- Substitute $y = 2k - 1$ into $5y = k + 4$: $5(2k - 1) = k + 4$.
- Distribute 5: $10k - 5 = k + 4$.
- Subtract k from both sides: $9k - 5 = 4$.
- Add 5 to both sides: $9k = 9$.
- Divide both sides by 9: $k = 1$.
- Find the solution: $y = 2(1) - 1 = 1$ and $x = 2y = 2$.
- Check the first equation: $2(2) + 1 = 5$ and $k + 4 = 5$. Check the second: $2 - 1 = 1$ and $2k - 1 = 1$. Both true.
- The value $k = 1$ is option (D).
- So the answer is (D) 1 .

Answer: (D)

27.

SAT CHALLENGE · DESMOS

The graph shows two lines in the xy -plane. The two lines and the x -axis form a triangle (shaded). What is the area of the triangle?



1. Given (read from the existing steps): the first line passes through $(-3, 0)$ and $(0, 3)$, the second passes through $(0, 6)$ and $(4, 0)$, and the triangle is bounded by the two lines and the x -axis. Asked: the area.
2. Method: find the intersection of the two lines (the top vertex), then use base along the x -axis and the vertex height in $\frac{1}{2} \cdot \text{base} \cdot \text{height}$.
3. First line: slope $\frac{3-0}{0-(-3)} = 1$, intercept 3, so $y = x + 3$.
4. Second line: slope $\frac{0-6}{4-0} = -\frac{3}{2}$, intercept 6, so $y = -\frac{3}{2}x + 6$.
5. Set equal: $x + 3 = -\frac{3}{2}x + 6$.
6. Add $\frac{3}{2}x$ to both sides: $\frac{5}{2}x + 3 = 6$.
7. Subtract 3: $\frac{5}{2}x = 3$. Multiply by $\frac{2}{5}$: $x = \frac{6}{5}$.
8. Substitute into $y = x + 3$: $y = \frac{6}{5} + 3 = \frac{6}{5} + \frac{15}{5} = \frac{21}{5}$.
9. Check in the second line: $-\frac{3}{2} \cdot \frac{6}{5} + 6 = -\frac{9}{5} + \frac{30}{5} = \frac{21}{5}$.
10. The lines meet the x -axis where $y = 0$: the first at $x = -3$, and the second at $0 = -\frac{3}{2}x + 6$, so $x = 4$.
11. Base along the x -axis: $4 - (-3) = 7$.
12. Height: the y -coordinate of the top vertex, $\frac{21}{5}$.
13. Area: $\frac{1}{2} \cdot 7 \cdot \frac{21}{5} = \frac{147}{10}$.
14. Convert to a decimal: $147 \div 10 = 14.7$.
15. So the answer is 14.7.

Answer: 14.7

28.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$6x - 9y = 15$$

$$2x + by = c$$

In the system of equations above, b and c are integers and $1 \leq c \leq 10$. For how many ordered pairs (b, c) does the system have no solution?

- Given: $6x - 9y = 15$ and $2x + by = c$, where b and c are integers and $1 \leq c \leq 10$. Asked: how many ordered pairs (b, c) give no solution.
- Method: no solution means parallel, different lines. First find the b that makes the lines parallel, then count the c values that make them different.
- Set the ratio of x -coefficients equal to the ratio of y -coefficients: $\frac{2}{6} = \frac{b}{-9}$.
- Simplify: $\frac{1}{3} = \frac{b}{-9}$. Multiply both sides by -9 : $b = -3$.
- Any other b gives different slopes, so exactly one solution. Only $b = -3$ can give no solution.
- With $b = -3$ the second equation is $2x - 3y = c$. Multiply by 3: $6x - 9y = 3c$.
- Compare with $6x - 9y = 15$: the lines are the same exactly when $3c = 15$, that is $c = 5$.
- For $c \neq 5$ the expression $6x - 9y$ would have to equal both 15 and $3c$, which differ, so there is no solution.
- Check $c = 5$: $2x - 3y = 5$ times 3 is $6x - 9y = 15$, the same line, so $c = 5$ is excluded.
- The allowed values of c are $1, 2, \dots, 10$, which is 10 values. Remove $c = 5$: $c \in \{1, 2, 3, 4, 6, 7, 8, 9, 10\}$, which is 9 values.
- Each of these gives one pair $(-3, c)$.
- There are 9 ordered pairs.
- So the answer is 9.

Answer: 9

29.

SAT CHALLENGE · CALCULATOR

Marcus jog for h hours at 8 miles per hour. He then walks for twice as many hours as he jogged, at 3 miles per hour, and finally bikes at 12 miles per hour for 30 minutes less than the time he jogged. The total distance of the trip is 46 miles. How many minutes did the whole trip take?

- (A) 450 ✓ (B) 480
(C) 7.5 (D) 420

- Given: jog for h hours at 8 miles per hour; walk for $2h$ hours at 3 miles per hour; bike at 12 miles per hour for $\left(h - \frac{1}{2}\right)$ hours; total distance 46 miles. Asked: the total trip time in minutes.
- Method: distance equals speed times time. Find each distance in terms of h , add to make 46, solve for h .
- Jog distance: $8h$.
- Walk distance: $3(2h) = 6h$.
- Bike distance: $12\left(h - \frac{1}{2}\right) = 12h - 6$.
- Total distance: $8h + 6h + (12h - 6) = 46$.
- Combine like terms ($8h + 6h + 12h = 26h$): $26h - 6 = 46$.
- Add 6 to both sides: $26h = 52$.
- Divide both sides by 26: $h = 2$.
- Times: jog 2 hours, walk $2h = 4$ hours, bike $2 - \frac{1}{2} = \frac{3}{2}$ hours. Total: $2 + 4 + 1.5 = 7.5$ hours.
- Convert to minutes: $7.5 \times 60 = 450$.
- Check distance: jog $8 \cdot 2 = 16$, walk $3 \cdot 4 = 12$, bike $12 \cdot 1.5 = 18$, and $16 + 12 + 18 = 46$.
- The value 450 is option (A).
- So the answer is (A) 450.

Answer: (A)

30.

SAT CHALLENGE · NO CALCULATOR NEEDED

A box contains only red, green and blue marbles. The ratio of red marbles to green marbles is 2 to 3, and the ratio of green marbles to blue marbles is 4 to 5. After 9 red marbles are removed and 9 blue marbles are added, there are exactly three times as many blue marbles as red marbles. How many marbles were in the box originally?

1. Given: red : green = 2 : 3 and green : blue = 4 : 5. After removing 9 red and adding 9 blue, blue is 3 times red. Asked: the original number of marbles.
2. Method: combine the two ratios using a common green amount, name the counts with one multiplier k , and use the after-change condition.
3. Scale red : green = 2 : 3 by 4: 8 : 12.
4. Scale green : blue = 4 : 5 by 3: 12 : 15.
5. Now green is 12 in both, so red : green : blue = 8 : 12 : 15. Let the counts be $8k$, $12k$, $15k$.
6. After the change: red = $8k - 9$ and blue = $15k + 9$.
7. Blue is three times red: $15k + 9 = 3(8k - 9)$.
8. Distribute 3: $15k + 9 = 24k - 27$.
9. Subtract $15k$ from both sides: $9 = 9k - 27$.
10. Add 27 to both sides: $36 = 9k$.
11. Divide by 9: $k = 4$.
12. Original counts: red $8(4) = 32$, green $12(4) = 48$, blue $15(4) = 60$. Total = $32 + 48 + 60 = 140$.
13. Check ratios: $32 : 48 = 2 : 3$ and $48 : 60 = 4 : 5$. After the change red is $32 - 9 = 23$ and blue is $60 + 9 = 69$, and $3 \cdot 23 = 69$.
14. So the answer is 140.

Answer: 140**31.**

SAT CHALLENGE · CALCULATOR

A science department orders microscopes at \$240 each and slide kits at \$15 each. The order must include at least 6 slide kits for every microscope, the total cost may not exceed \$2,500, and the storage cabinet holds at most 46 items (microscopes and kits combined). What is the greatest number of microscopes the department can order?

1. Given: microscope 240 dollars each, slide kit 15 dollars each. Conditions: at least 6 kits per microscope, total cost at most 2500 dollars, at most 46 items. Asked: the greatest number of microscopes.
2. Method: write the three inequalities, then find the smallest number of kits allowed for each m .
3. Let m be microscopes and k be kits: $k \geq 6m$, $240m + 15k \leq 2500$, $m + k \leq 46$.
4. Both the cost and the item count increase when k increases, so the easiest case for a given m is the smallest allowed k , namely $k = 6m$.
5. Cost with $k = 6m$: $240m + 15(6m) = 240m + 90m = 330m \leq 2500$.
6. Divide by 330: $m \leq \frac{2500}{330} \approx 7.58$, so $m \leq 7$.
7. Items with $k = 6m$: $m + 6m = 7m \leq 46$.
8. Divide by 7: $m \leq \frac{46}{7} \approx 6.57$, so $m \leq 6$.
9. Both conditions must hold, so $m \leq 6$.
10. Check $m = 6$, $k = 36$: kits $36 \geq 6(6) = 36$, cost $240(6) + 15(36) = 1440 + 540 = 1980 \leq 2500$, items $6 + 36 = 42 \leq 46$. All hold.
11. Check $m = 7$: at least 42 kits are needed, so items $\geq 7 + 42 = 49$, and $49 > 46$.
12. So the greatest number of microscopes is 6.
13. So the answer is 6.

Answer: 6

32.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$3x + 2y = 12$$

$$9x + 6y = 36$$

For the system above, which of the following ordered pairs (x, y) , where t is any real number, is a solution to the system?

(A) $(4 - 2t, -3t)$

(B) $(2t + 4, -3t)$ ✓

(C) $(2t + 4, 3t)$

(D) $(3t + 4, -2t)$

- Given: $3x + 2y = 12$ and $9x + 6y = 36$. Asked: which ordered pair in terms of t is a solution for every t .
- Method: compare the equations. Multiplying the first by 3 gives $9x + 6y = 36$, the second equation, so both describe the same line.
- So a pair is a solution exactly when it satisfies $3x + 2y = 12$. We substitute each option and look for the value 12 for every t .
- (A) $x = 4 - 2t, y = -3t$: $3(4 - 2t) + 2(-3t) = 12 - 6t - 6t = 12 - 12t$, which changes with t .
- (B) $x = 2t + 4, y = -3t$: $3(2t + 4) + 2(-3t) = 6t + 12 - 6t = 12$. The t terms cancel and the value is always 12.
- (C) $x = 2t + 4, y = 3t$: $3(2t + 4) + 2(3t) = 6t + 12 + 6t = 12t + 12$, which changes with t .
- (D) $x = 3t + 4, y = -2t$: $3(3t + 4) + 2(-2t) = 9t + 12 - 4t = 5t + 12$, which changes with t .
- Check (B) in the second equation: $9(2t + 4) + 6(-3t) = 18t + 36 - 18t = 36$, true for every t .
- Only (B) satisfies the system for every t .
- So the answer is (B) $(2t + 4, -3t)$.

Answer: (B)

33.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$6x - 2(y - 5) = 2x + 3y + 1$$

$$ax + 20y = b$$

In the system above, a and b are constants. If the system has infinitely many solutions, what is the value of $\frac{b}{a}$?

- Given: $6x - 2(y - 5) = 2x + 3y + 1$ and $ax + 20y = b$, with infinitely many solutions. Asked: the value of $\frac{b}{a}$.
- Method: simplify the first equation to $Ax + By = C$ form. Infinitely many solutions means the second equation is a constant multiple of the first.
- Distribute -2 : $6x - 2y + 10 = 2x + 3y + 1$.
- Subtract $2x$ and $3y$ from both sides: $4x - 5y + 10 = 1$.
- Subtract 10 from both sides: $4x - 5y = -9$.
- Let the second equation be k times this one: $4kx - 5ky = -9k$, so it must equal $ax + 20y = b$.
- Match the y -coefficients: $-5k = 20$. Divide by -5 : $k = -4$.
- Then $a = 4k = 4(-4) = -16$.
- And $b = -9k = -9(-4) = 36$.
- Compute $\frac{b}{a} = \frac{36}{-16}$. Divide the numerator and denominator by 4: $-\frac{9}{4}$.
- Check: $-16x + 20y = 36$ divided by -4 gives $4x - 5y = -9$, the simplified first equation.
- So the answer is $-\frac{9}{4}$ (that is, $-9/4$).

Answer: $-9/4$

34.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$ax + ay = 15$$

$$3ax - 2ay = 20$$

In the system of equations above, a is a nonzero constant. The ordered pair $(2, y)$ is a solution to the system. What is the value of a ?

1. Given: $ax + ay = 15$ and $3ax - 2ay = 20$, with $a \neq 0$, and $(2, y)$ is a solution. Asked: the value of a .
2. Method: substitute $x = 2$, treat the product ay as one unknown, and use elimination.
3. First equation with $x = 2$: $a(2) + ay = 15$, which is $2a + ay = 15$.
4. Second equation with $x = 2$: $3a(2) - 2ay = 20$, which is $6a - 2ay = 20$.
5. Let $P = ay$. The equations become $2a + P = 15$ and $6a - 2P = 20$.
6. Multiply the first by 2: $4a + 2P = 30$.
7. Add it to the second equation: $(4a + 6a) + (2P - 2P) = 30 + 20$, so $10a = 50$.
8. Divide by 10: $a = 5$.
9. Find P : $2(5) + P = 15$, so $P = 5$. Since $P = ay$, $5y = 5$ and $y = 1$.
10. Check: $5(2) + 5(1) = 15$ and $3(5)(2) - 2(5)(1) = 30 - 10 = 20$, both true.
11. So the answer is 5.

Answer: 5

35.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$\frac{x}{6} + \frac{y}{4} = 2$$

Which of the following equations, when paired with the equation above, forms a system of two linear equations in x and y that has no solution?

(A) $4x - 6y = 40$

(B) $4x + 6y = 48$

(C) $4x + 6y = 40$ ✓

(D) $6x + 4y = 48$

1. Given: $\frac{x}{6} + \frac{y}{4} = 2$. Asked: which option, paired with it, gives a system with no solution.
2. Method: clear the fractions, then compare. No solution means parallel, different lines.
3. Multiply every term by 12: $12 \cdot \frac{x}{6} = 2x$, $12 \cdot \frac{y}{4} = 3y$, $12 \cdot 2 = 24$. The equation is $2x + 3y = 24$.
4. Doubling it gives $4x + 6y = 48$. The slope of $2x + 3y = 24$ is found from $3y = -2x + 24$, so $y = -\frac{2}{3}x + 8$.
5. (A) $4x - 6y = 40$: $-6y = -4x + 40$, so $y = \frac{2}{3}x - \frac{20}{3}$. The slope $\frac{2}{3}$ differs from $-\frac{2}{3}$, so one solution.
6. (B) $4x + 6y = 48$ is exactly $2(2x + 3y = 24)$, the same line, so infinitely many solutions.
7. (C) $4x + 6y = 40$: the left side is $2(2x + 3y)$, and the given equation says $2x + 3y = 24$, so it would equal 48. Option (C) says 40, and $48 \neq 40$, so no pair satisfies both. No solution.
8. (D) $6x + 4y = 48$: $4y = -6x + 48$, so $y = -\frac{3}{2}x + 12$. The slope $-\frac{3}{2}$ differs from $-\frac{2}{3}$, so one solution.
9. Check (C) in slope-intercept form: $6y = -4x + 40$ gives $y = -\frac{2}{3}x + \frac{20}{3}$. Same slope as the given line, and the intercepts $\frac{20}{3}$ and 8 differ, so the lines are parallel and different.
10. So the answer is (C) $4x + 6y = 40$.

Answer: (C)

36.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$3x + ky = 12$$

$$21x + 14y = 6m$$

In the system of equations above, k and m are constants. If the system has infinitely many solutions (x, y) , what is the value of km ?

1. Given: $3x + ky = 12$ and $21x + 14y = 6m$, with infinitely many solutions. Asked: the value of km .
2. Method: infinitely many solutions means the second equation is the first equation multiplied by a constant.
3. Compare the x -coefficients: $21 = 7 \cdot 3$, so the constant is 7.
4. Multiply every term of $3x + ky = 12$ by 7: $7(3x) = 21x$, $7(ky) = 7ky$, $7(12) = 84$.
5. The result is $21x + 7ky = 84$.
6. Match the y -coefficients with $21x + 14y = 6m$: $7k = 14$. Divide by 7: $k = 2$.
7. Match the constants: $84 = 6m$. Divide by 6: $m = 14$.
8. Compute $km = 2 \cdot 14 = 28$.
9. Check: with $k = 2$ the first equation is $3x + 2y = 12$. Multiplying by 7 gives $21x + 14y = 84$, and $6m = 6(14) = 84$.
10. Both equations are the same line.
11. So the answer is 28.

Answer: 28